

CTS SRE

Automation for Toil Reduction

EMPOWERING High Performance
Technology Teams

Agenda for the Day

- Understanding Toil & Its Impact on Operations
- Identifying High-Impact Toil Areas
- Automation Strategies Using Terraform & Ansible
- Self-Healing Mechanisms for System Resilience
- Automating Environment Provisioning & Deployments
- Standardizing Operational Task Automation
- Automated Recovery Procedures for Critical Services
- GenAI-Driven Log Analysis for Predictive Issue Resolution
- Case study presentation review.
- Summary & Key Takeaways

Understanding Toil & Its Impact on Operations

- **What is Toil?**
 - Manual, repetitive, and operationally expensive work that does not scale.
 - Examples: **Manually restarting crashed services, managing infrastructure manually, and handling repetitive alerts.**
- **Why Toil Reduction Matters?**
 - Increases developer productivity by reducing human intervention.
 - Enhances reliability and consistency of system operations.
 - Accelerates incident resolution through automation.

Identifying High-Impact Toil Areas

Common Toil Areas in Operations:

Manual service restarts and failure resolutions.

Infrastructure provisioning delays for new environments.

Alert fatigue due to repetitive monitoring notifications.

Manual deployments with frequent configuration errors.

How to Prioritize Automation? - **Jira EPIC/Stories and put in backlog**

- Identify **repetitive and error-prone** tasks. - **Discovery story in Jira**
- Focus on **high-frequency operational activities**.
- Ensure **automated solutions improve efficiency** without adding complexity.

Automation Strategies Using Terraform & Ansible

Terraform (Infrastructure as Code - IaC): Cloud agnostic/Multi cloud

Automates cloud infrastructure provisioning.

Provides scalable and consistent infrastructure management.

Ansible (Configuration Management):

Automates software deployments & system configurations.- JDK, Security software inside VM 1000's VMs

Ensures idempotency—executing scripts without causing inconsistencies.

 **Example:**

Terraform: Automatically provisioning EC2 instances, VPCs, security groups.

Ansible: Configuring servers to install & restart services dynamically./Network automation firewalls,routers, switches etc

Self-Healing Mechanisms for System Resilience

What are Self-Healing Systems?

Automatically detect failures and **recover without human intervention.**

Key Self-Healing Approaches:

Automated container restarts when health checks fail.

Auto-scaling groups to replace unhealthy instances.

Database failover mechanisms for high availability.

✂ **Example: Using Kubernetes Liveness & Readiness Probes** to restart unhealthy pods.

Automating Environment Provisioning & Deployments



Challenges in Manual Provisioning:

Delays in setting up **development, staging, and production environments**.
- 3 clusters

Risk of **configuration drift** between different environments.



Automation Benefits:

CI/CD Pipelines to deploy applications without manual intervention.

Automated environment setup with pre-configured settings.

Terraform & Ansible integration for provisioning & configuration consistency.



Example: Using Terraform to provision a **Kubernetes cluster with autoscaling enabled**. -Dev, stag, prod with multi account strategy

Standardizing Operational Task Automation



Repetitive Operational Tasks:

Monitoring & Alert Management (threshold-based alerts, anomaly detection).

Backup & Restoration Procedures (scheduled snapshots, failover setups).

Routine Maintenance & Patch Management (automated security updates).



Standardization with Templates:

Define **IaC modules** for common infrastructure patterns.

Create **pre-built Ansible playbooks** for frequent operations.

Automate **alert correlation & incident tagging**.

GenAI-Driven Log Analysis for Predictive Issue Resolution

How GenAI Can Improve Operations?

- Analyze large volumes of logs & metrics for hidden failure patterns.
- Automatically detect anomalies & recommend fixes.
- Reduce MTTR (Mean Time to Resolution) by prioritizing alerts based on impact.

GenAI-Driven Incident Resolution:

- Automated Root Cause Analysis (RCA) using AI models.
- Self-learning systems that suggest optimizations over time.
-  Example: AI model detects recurring API failures and automatically applies retry logic.

Breakout

Summary & Key Takeaways

- ✓ **Toil Reduction** improves productivity and system resilience. ✓ **Infrastructure Automation** using Terraform & Ansible eliminates manual interventions. ✓ **Self-Healing Mechanisms** reduce downtime and improve service reliability. ✓ **Standardized Automation Templates** streamline operational efficiency. ✓ **GenAI-driven log analysis** enables predictive issue resolution and RCA automation.

Introduction to case studies

Expectation from case study



Participants are expected to work on the deliverables:

- Define **SLI/SLO metrics** and create monitoring dashboards.
- Automate cloud infrastructure provisioning and deployments.
- Run **chaos experiments** and document resilience gaps.
- Conduct **root cause analysis (RCA)** and propose improvements.

Thank You!
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